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Cristopher Soto

Data Engineer

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EXPERIENCE

Data Engineer

NYCSBUS

Jun 2022 - Present

- **Architected and developed** end-to-end **data pipelines** that ingested, transformed, and delivered operational data from external sources into internal business applications, reporting systems, and stakeholder-facing tools.
- **Built ETL workflows** for web-scraped and operational datasets, automating data collection, transformation, and delivery into **Google Sheets** and other downstream systems used for business reporting and decision-making.
- **Engineered** a real-time **parent notification platform** for school bus proximity alerts using **AWS Lambda**, **DynamoDB**, **AWS Pinpoint**, and a custom geocoding API powered by **OpenStreetMap** and deployed behind **Amazon EC2 Auto Scaling Groups**.
- **Designed and implemented geospatial data pipelines** in **Apache Airflow** to capture and process school, student, and depot arrival events for school buses, enabling location-based operational tracking and analytics.
- **Developed** event-driven **AWS data pipelines** in which inbound emails received through **Amazon SES** triggered **Amazon S3** and **AWS Lambda** workflows to parse, transform, and load structured data into **PostgreSQL** on **Amazon RDS**.
- **Built** event-driven, auto-scaling **OCR data pipelines** using **AWS Textract** and **Tesseract** to process user-submitted documents, extract structured data, and support downstream operational and reporting workflows.
- **Engineered** scalable document-processing workflows that dynamically adjusted to user input volume, leveraging **serverless** and cloud-based architectures to improve reliability, efficiency, and throughput.
- **Built** full-stack internal applications using **AWS Amplify**, **AWS CloudFormation**, and **Amazon Cognito**, delivering secure cloud-based tools with authentication and user management for business operations.
- **Led the development** of a custom **CRM** and **call center platform** on **Amazon Connect**, provisioned through **AWS CloudFormation** and integrated with proprietary web tools to support customer service and operational workflows.
- **Created** a custom **web interface** for call center agents using **AWS Amplify** and backend integrations with **Amazon Connect APIs**, improving agent productivity and streamlining case management processes.
- **Implemented** real-time and near-real-time **streaming data pipelines** using **Amazon Kinesis**, **Amazon S3**, **AWS Lambda**, and **PostgreSQL** to capture, process, and store call center and operational activity data.
- **Designed and modeled** an enterprise **data warehouse** on **Amazon Aurora PostgreSQL**, including multiple schemas and **star-schema dimensional models** to support analytics, reporting, and cross-functional business intelligence use cases.
- **Built** custom **monitoring dashboards** in **Amazon CloudWatch** and developed multiple **alarm and notification systems** using **Amazon SNS** to proactively detect system failures, pipeline issues, and operational incidents.
- **Improved** platform reliability and operational visibility by implementing automated alerting, centralized monitoring, and failure notification workflows across cloud-based systems and data pipelines.
- **Collaborated** closely with business stakeholders to gather requirements, identify operational pain points, and design scalable **cloud-based solutions** aligned with business and technical needs.
- **Drove automation and modernization** efforts by replacing manual workflows with scalable, maintainable, and event-driven architectures across the **AWS** ecosystem.

EDUCATION AND COURSEWORK

AWS Certified Data Engineer – Associate , Score: 757	2026
AWS Certified Developer – Associate , Score: 810	2022
Coursework toward B.S. in Computer Science , Hunter College, CUNY (GPA: 3.8)	2020 – 2023

SKILLS

Programming Languages	Python, SQL, JavaScript,
Cloud Platforms & Services	AWS Lambda, AWS Amplify, Amazon Connect, AWS CloudFormation, Amazon Cognito, Amazon Kinesis, Amazon S3, Amazon SES, Amazon SNS, Amazon CloudWatch, AWS Textract, AWS Glue, Amazon EMR
Frameworks & Tools	React, Apache Airflow, Tesseract, Google Sheets
Databases & Warehousing	PostgreSQL, Amazon Aurora, DynamoDB, Amazon Redshift

PROJECTS

AWS ETL Pipeline

- Built an **ETL pipeline** with **Apache Airflow**, **Amazon S3**, **Boto3**, and **PostgreSQL** on **AWS RDS**.
- Developed and deployed an interactive **Dash** dashboard on **Heroku** using **Gunicorn**.

Name Querying and Data Visualization Web Application

- Built a **data visualization web application** to analyze **Social Security name data** from 1910 to the present.
- Created interactive **Plotly** visualizations, including a choropleth map and time-series chart, and deployed the app on **Heroku**.